AGENTX HACKATHON 2026 · TRACK PS-04

ReturnLens

Every return decision, backed by evidence.

Evidence-backed returns triage for independent online sellers.

When a customer submits a return video, ReturnLens extracts keyframes, cross-references order truth and shipping timestamps, verifies deterministic code policies, and gives the solo seller a 1-click decision console.

15–20m
Manual triage time eliminated

1-Click
Merchant human verification

24h
Prototype build feasibility

RETURNLENS SELLER CONSOLE — CASE #RT-8821



ORDER INFORMATION
Order #ORD-7729 · HEAD-WHT-02
Matte White · Price: ₹1,499
Delivered 4 days ago

EVIDENCE CITATION (STAGED)
Frame 00:07 (Staged)
Observed item casing is Midnight Black. Exact barcode unverified.

POLICY GATE: COLOR CONTRADICTION DETECTED
P-04 Return Window: PASS (Day 4 of 7) | P-03 Manifest Match: CONTRADICTION FLAGGED

STATUS: ESCALATED FOR SELLER DECISION

Review Evidence & Authorize

THE PROBLEM · SOLO SELLER BOTTLENECK

One return. Four disconnected sources.

Four systems. No traceable decision. · Without a shared record, the seller connects order facts, video, and policy manually from memory.

01 BUYER VIDEO & CLAIM

Buyer Video & Claim

Customer submits short unboxing clip and subjective text via chat.

02 STORE ORDER TRUTH

Store Order Truth

Shopify store catalog specifies SKU HEAD-WHT-02 (White, ₹1,499).

CRITICAL INFORMATION GAP

NO SHARED EVIDENCE RECORD

Today the seller opens 4 browser tabs and guesses. An erroneous refund burns inventory margin; an unfair denial destroys repeat customer trust. The gap is the absence of an auditable artifact tying order truth, video keyframes, and policy clauses together before money moves.

9.0% vs 15.14%

9.0% Return Fraud (NRF 2025, fraud only) | 15.14% Fraud + Policy Abuse (Appriss/Deloitte 2024)
Definitions differ: strict criminal fraud vs fraud combined with policy abuse.

03 COURIER TRACKING SCAN

Courier Tracking Scan

Shippo courier API records delivery scan 4 days ago.

04 STORE RETURN POLICY

Store Return Policy

7-day defect window and tag condition in store terms.

Fraud statistics note: 9.0% (NRF 2025, fraud only) vs 15.14% (Appriss/Deloitte 2024, fraud + abuse)

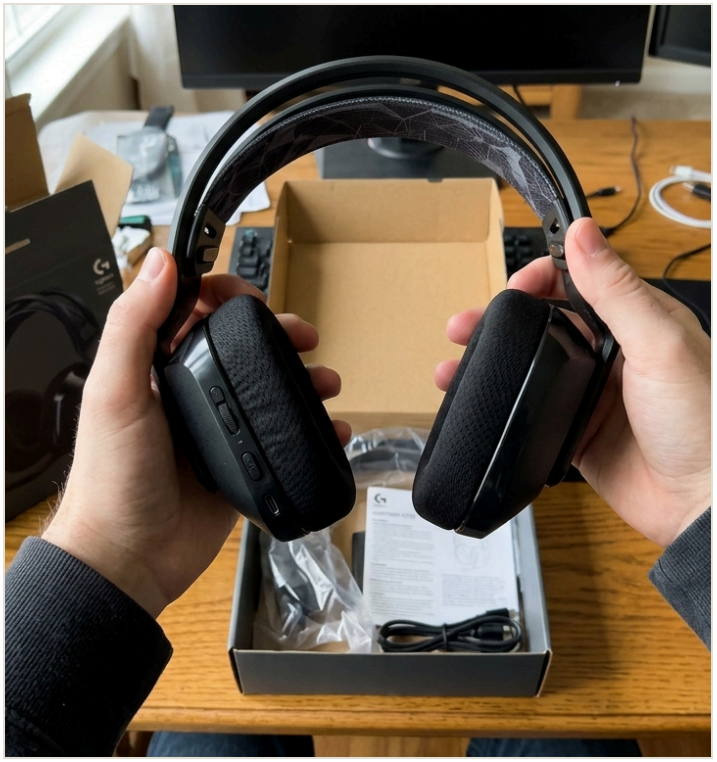
VISUAL EVIDENCE BENCHMARK · STAR DEMO

The order says white. The video shows black.

ReturnLens flags the contradiction — the seller decides.



STORE CATALOG TRUTH: HEAD-WHT-02 (WHITE)



UNBOXING KEYFRAME (STAGED): MIDNIGHT BLACK

Visual Grounding Honesty: Keyframe resolution reveals black casing. Exact SKU barcode is unreadable in this frame; ReturnLens factually records 'Observed Color: Midnight Black · Exact SKU: Not verified' and cites frame 00:07.

DETERMINISTIC POLICY GATE

P-03: COLOR CONTRADICTION

SELLER REVIEW REQUIRED
Observed: Midnight Black · Exact SKU: Not verified in frame

Seller Resolution Options:

- 1. Request unboxing video with packaging label / serial visible
- 2. Reject return under policy (item does not match order record)
- 3. Override & approve if seller acknowledges known warehouse mis-shipment

Financial Safety Ceiling:
`0 < amount <= min(₹4,999, remaining_refundable, order_paid)`
No auto-denial · No refund released without seller click

SYSTEM ARCHITECTURE · DYNAMIC AGENTIC GOVERNANCE

Models observe. Code verifies. Seller authorizes.

Dynamic action selection and stateful resumption under deterministic human authorization.

01 PLAN	02 RETRIEVE	03 EVALUATE	04 AUDIT	05 AUTHORIZE	06 RECORD
Plan	**Retrieve**	**Evaluate**	**Audit**	**Authorize**	**Record**
MODEL	CODE	CODE	MODEL	HUMAN	CODE

CASE C · DYNAMIC STATEFUL CLARIFICATION LOOP

Observe → Decide → Act → Wait → Resume

Dark Frames → ask_for_evidence() → **WAITING_FOR_EVIDENCE** → Customer Uploads → Resume Case → Seller Review

When lighting is insufficient, the agent halts standard execution, prompts a targeted checklist, pauses state in WAITING_FOR_EVIDENCE, and seamlessly resumes context upon media receipt (1 retry limit protects customer experience).

SECONDARY PS-04 ROUTE: ORDER STATUS FAST PATH

Zero-Vision Direct Route

"Where is my order?" → Query Store & Shippo APIs → Delivery Scan & Time

Routine order queries take a direct tool route without calling heavy multimodal models. Escalates to human seller only if courier scan shows abnormal transit delay.

- **Model Calls:** 3 base LLM calls per triage case (Planner, Vision, Critic) plus retries.
- **Critic Duty:** Independent prompt audits cited frames; notes correlated vision limits.
- **Deterministic Policy:** Rules P-01 to P-08 executed in pure Python code, never prompts.
- **Safety Ceiling:** Backend checks block payment tools; 1-click human authorization strictly enforced.

DEMO SCENARIOS · CONTROLLED PROTOTYPE BEHAVIORS

Four evidence conditions. Four controlled actions.

Expected prototype behaviors across staged benchmark cases. Not measured production metrics.

CASE 01 · CONSISTENT	Staged casing crack verified in unboxing frame 00:04 · Manifest & return window match.	→	**PREPARE SELLER REVIEW** Agent prepares mock shipping label & drafts refund in pending_return_receipt. Money releases only upon warehouse receipt.
CASE 02 · CONTRADICTION	White ordered (HEAD-WHT-02) vs Black unboxed in Frame 00:07 · Exact SKU unverified.	→	**ESCALATE TO SELLER** Policy P-03 blocks automation. Inhibits label and refund. Seller selects 1 of 3 concrete resolution options.
CASE 03 · UNCLEAR	4-second dark video, brightness 28/255, barcode obscured · Insufficient visual evidence.	→	**ASK AND RESUME** Agent calls ask_for_evidence(); transitions ticket to WAITING_FOR_EVIDENCE; resumes context on upload (1 retry limit).
CASE 04 · SEQUENCE	Staged continuous take from sealed parcel opening to product reveal · Sequence inspected.	→	**INSPECT CONTINUITY** Expected prototype behavior: inspect sequence continuity and flag missing transitions for seller review.

A credible prototype in 24 hours.

Strict scope boundaries and single-threaded ownership make the prototype fully buildable in one day.

00h – 06h · FOUNDATION

Mock APIs & Agent State

Person A (Backend)

LangGraph state machine, mock Shopify/Shippo fixtures, order status route, state serialization.

06h – 12h · EVIDENCE

Evidence Pipeline & Policy

Person B (Vision & UI)

FFmpeg keyframe extraction, vision schema, deterministic rules P-01 to P-08, critic prompt.

12h – 18h · INTEGRATION

State Flow & Scenarios

Person C (Integration) with A/B

Clarification/resume loop, seller console UI, wiring 4 staged demo cases (A–D).

18h – 24h · VALIDATION

Testing & Submission

All 3 Members

Authorization boundary tests, edge cases, fallback validation, demo recording, README.

MUST DELIVER (CORE PROTOTYPE)

- Deterministic policy engine (P-01 to P-08)
- Evidence-pack keyframe citations
- Fail-closed seller authorization gate
- Structured JSON audit trail log
- Stateful clarification/resume loop
- 4 reproducible demo scenarios

CUT FIRST IF BEHIND (DEFERRED EXPANSION)

- Live carrier webhooks (keep mock courier)
- Multi-item return parsing (keep 1 item/return)
- Video OCR barcode scanner (keep visual color)
- Live payment gateways (keep draft refunds)
- Merchant analytics dashboard